

Shijie Wang

PhD Candidate in Computer Science | LLMs · RAG · Recommender Systems
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sjay-wang.github.io | Google Scholar: 2,170+ citations, h-index: 11

Summary

PhD candidate in Computer Science, expected to graduate in Fall 2026, specializing in LLM-based recommendation, retrieval-augmented generation, and trustworthy recommender systems. First author of papers at KDD, ACL, and TOIS, with 2,170+ Google Scholar citations. Industry research experience at Baidu Search Science Team, with expertise in knowledge-grounded recommendation, multi-agent LLM systems, and ranking evaluation.

Education and Visiting Experience

PhD Candidate, Department of Computing, The Hong Kong Polytechnic University Advisors: Prof. Wenqi Fan and Prof. Qing Li	Hong Kong 08/2022–Present
Visiting Research Scholar, Institute of Data Science, National University of Singapore Advisor: Prof. See-Kiong Ng	Singapore 01/2026–08/2026
BSc in Information and Computing Science, University of Liverpool Grade: First-Class Honours	Liverpool, UK 09/2018–07/2022

Work Experience

Baidu Research Intern	06/2024–01/2025 Search Science Team
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Mentors: Dr. Dawei Yin and Dr. Shuaiqiang Wang

- Developed **K-RagRec**, a knowledge graph retrieval-augmented framework that grounds LLM-based recommendation with structured, up-to-date knowledge, improving Recall by **41.6% on average** and resulting in a first-author paper at **ACL 2025 Main**.
- Designed retrieval and knowledge-integration strategies that combine user–item interaction signals with relevant KG structures to improve recommendation quality and mitigate LLM hallucination.
- Built an end-to-end experimentation pipeline covering knowledge retrieval, candidate construction, top- K ranking evaluation, baseline comparison, and ablation studies.
- Collaborated with Baidu Search Science researchers to formulate industrially relevant recommendation problems and translate research ideas into reproducible empirical studies.

Selected Research Contributions

LLM/RAG-enhanced Recommender Systems

- K-RagRec (ACL 2025):** Proposed a KG-based retrieval-augmented recommendation framework that grounds LLMs with high-quality structural knowledge to mitigate hallucination and improve top- K recommendation.
- MixRAGRec (KDD 2026):** Extended KG-RAG recommendation with a cooperative multi-agent framework that adaptively routes queries to retrieval experts at different graph granularities and jointly optimizes retrieval, alignment, and recommendation.
- RecBench+ (WSDM 2026):** Developed a benchmark that exposes the reasoning challenges of personalized LLM recommendation assistants through realistic queries with hard constraints, soft preferences, and varying difficulties.
- ReRec (ACL 2026):** Addressed these reasoning challenges through reinforcement fine-tuning with dual-graph reward shaping, reasoning-aware advantage estimation, and curriculum learning.
- Atomic Intent Reasoning (KDD Industry 2026):** Introduced an industrial cross-domain recommendation framework that performs LLM semantic reasoning offline and dynamically retrieves and composes atomic user intents online, achieving approximately 400× faster inference and a 3.446% GMV gain in large-scale online A/B testing.

Trustworthy and Safe Recommender Systems

- MultiAttack (TOIS 2024):** Developed a multi-agent reinforcement learning framework that coordinates malicious user profiles and cross-community social relations to evaluate black-box social recommender robustness.
- CheatAgent (KDD 2024):** Extended black-box recommendation security research to LLM-empowered systems through an LLM attack agent that identifies high-impact insertion positions and iteratively generates adversarial textual perturbations.

Selected Publications

* indicates equal contribution.

- [1] **Shijie Wang**, Chengyi Liu, Yajuan Ding, Shanru Lin, See-Kiong Ng, Xu Xin, Wenqi Fan. “Mixture-of-Experts Knowledge Graph Retrieval-Augmented Generation for Multi-Agent LLM-based Recommendation.” **KDD 2026 (CCF-A)**.
- [2] **Shijie Wang**, Wenqi Fan, Yue Feng, Xinyu Ma, Shuaiqiang Wang, Dawei Yin. “Knowledge Graph Retrieval-Augmented Generation for LLM-based Recommendation.” **ACL Main 2025 (CCF-A)**.
- [3] **Shijie Wang**, Wenqi Fan, Xiao-yong Wei, Xiaowei Mei, Shanru Lin, Qing Li. “Multi-agent Attacks for Black-box Social Recommendations.” **TOIS 2024 (CCF-A)**.
- [4] **Shijie Wang**, Jiani Huang, Wenqi Fan, Zhikai Chen, Yu Song, Wenzhuo Tang, Haitao Mao, Hui Liu, Xiaorui Liu, Dawei Yin, Qing Li. “Graph Machine Learning in the Era of Large Language Models (LLMs).” **TIST 2025**.
- [5] Liang-bo Ning*, **Shijie Wang***, Wenqi Fan, Qing Li, Xu Xin, Hao Chen, Feiran Huang. “CheatAgent: Attacking LLM-Empowered Recommender Systems via LLM Agent.” **KDD 2024 (CCF-A)**.
- [6] Jiani Huang, **Shijie Wang**, Liang-bo Ning, Wenqi Fan, Qing Li. “ReRec: Reasoning-Augmented LLM-based Recommendation Assistant via Reinforcement Fine-tuning.” **ACL Main 2026 (CCF-A)**.
- [7] Jiani Huang, **Shijie Wang**, Liang-bo Ning, Wenqi Fan, Shuaiqiang Wang, Dawei Yin, Qing Li. “Towards Next-Generation Recommender Systems: A Benchmark for Personalized Recommendation Assistant with LLMs.” **WSDM 2026 (CCF-B)**.
- [8] Zhuohang Jiang, Yuxin Chen, **Shijie Wang**, Haohao Qu, Jindong Zhou, Wenqi Fan, Qing Li, Dongxu Liang, Jun Wang. “Atomic Intent Reasoning: Bringing LLM Semantics to Industrial Cross-Domain Recommendations.” **KDD Industry 2026 (CCF-A)**.
- [9] Chengyi Liu, Xiao Chen, **Shijie Wang**, Wenqi Fan, Qing Li. “Continuous-time Discrete-space Diffusion Model for Recommendation.” **WSDM 2026 (CCF-B)**.
- [10] Lin Wang, **Shijie Wang**, Sirui Huang, Qing Li. “Simplifying Graph Neural Kernels: from Stacking Layers to Collapsed Structure.” **ICWL 2026**.
- [11] Chengyi Liu, Jiahao Zhang, **Shijie Wang**, Wenqi Fan, Qing Li. “Score-based Generative Diffusion Models for Social Recommendations.” **TKDE 2025 (CCF-A)**.
- [12] Yuqi Zhang, Bin Guo, Nuo Li, Ying Zhang, **Shijie Wang**, Zhiwen Yu, Qing Li. “Tree-of-AdEditor: Heuristic Tree Reasoning for Automated Video Advertisement Editing with Large Language Model.” **IJCAI 2025 (CCF-A)**.
- [13] Yajuan Ding, Wenqi Fan, Liangbo Ning, **Shijie Wang**, Hengyun Li, Dawei Yin, Tat-Seng Chua, Qing Li. “A Survey on RAG Meets LLMs: Towards Retrieval-Augmented Large Language Models.” **KDD 2024 (CCF-A)**.
- [14] Jiahao Zhang, Lin Wang, **Shijie Wang**, Wenqi Fan. “Graph Unlearning with Efficient Partial Retraining.” **WWW PhD Symposium 2024**.
- [15] **Shijie Wang**, Shangbo Wang. “A Novel Multi-Agent Deep RL Approach for Traffic Signal Control.” **PerCOM Workshops 2023**.

Professional Service and Recognition

- **Outstanding Reviewer**, KDD 2025, top 10%.
- Reviewer for KDD, AAI, ICDM, WWW, IJCAI, MM, WSDM, CIKM, LOG, ACM TOIS, IEEE TKDE, IEEE TIFS, IEEE IoT-J, and ACM TKDD.
- Co-organized tutorials on RAG, LLM-based recommender systems, and trustworthy recommender systems at ICDE, KDD, IJCAI, and WWW.